

MMI714 Homework Assignment 2024/25 Fall

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I. INTRODUCTION

First, I generated 8 different datasets to observe how the algorithm's performance might vary depending on four factors: *means, covariances, number of clusters, and initialization method*.

I labeled these datasets based on the difficulty level of their means and covariances, referring to them as '**easy**' and '**difficult**'. 'Easy means' indicates that the cluster centers are far apart, making them easier for the algorithm to distinguish. In contrast, 'difficult means' suggests that the cluster centers are very close, which can make it challenging for the algorithm to correctly separate them.

Regarding covariances, 'easy covariances' refers to cases where the covariance matrix is an identity matrix, resulting in well-shaped, spherical clusters. On the other hand, 'difficult covariances' refer to covariance matrices that deviate from the identity matrix, often leading to elongated cluster shapes.

The datasets illustrate four levels of difficulty based on different combinations of means and covariances: (Fig. 1)

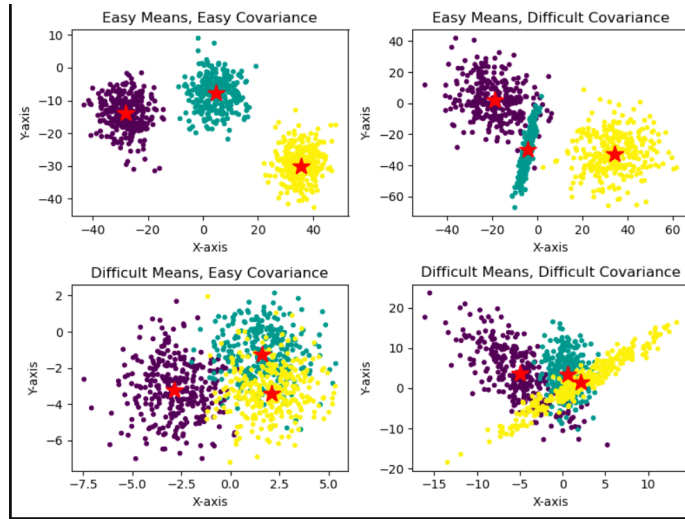


Fig. 1. Difficulty levels with respect to means and covariances.

Next, I applied these 4 difficulty levels to another parameter: *the number of clusters*. Since I wanted to create datasets with 2 different cluster counts (3 and 5), this resulted in a total of 8 unique datasets. (Fig. 2)

Although the data itself plays a significant role in determining whether the algorithm can successfully cluster it, the *initialization method* is also very important. If both the means (widely

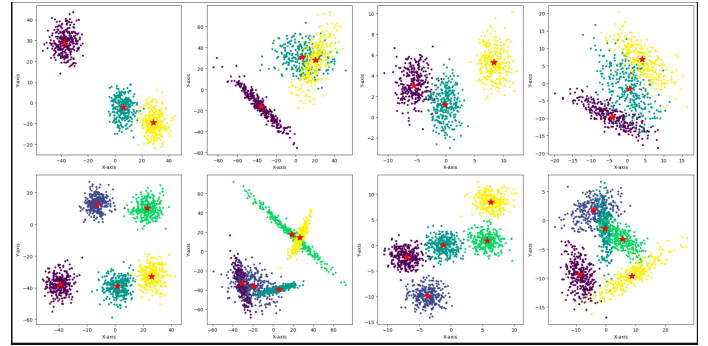


Fig. 2. Difficulty levels with respect to number of clusters, means and covariances.

separated) and covariances (identity matrix) are 'easy,' then the algorithm converges quickly, even when the number of clusters is increased. However, with poor initialization, the algorithm struggled to cluster the data correctly, even in the presence of easy means and covariances. This led me to develop a more effective initialization method, which consistently found the correct means.

To initialize the means, I selected n (the number of clusters) data points from the dataset, applying specific constraints to ensure better initialization. Similarly, I constructed the covariance matrix based on the data's variance patterns to enhance clustering accuracy. (Fig. 3)

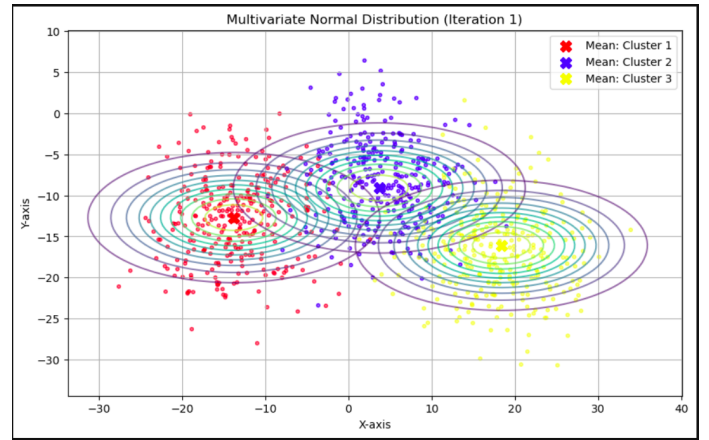


Fig. 3. Effective initialization method

The algorithm correctly clustered all the data points, even when there were 5 clusters and the data was challenging. (Fig. 4)

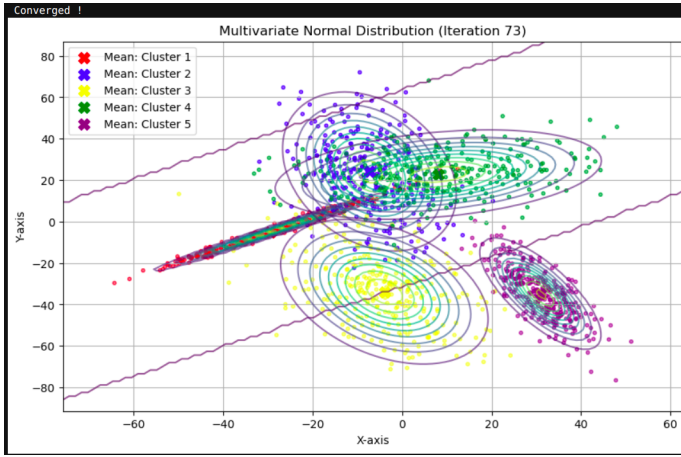


Fig. 4. Challenging data with 5 clusters.

Even when I made the means closer together, increasing the difficulty of the data, the algorithm was still able to cluster it correctly. (Fig. 5)

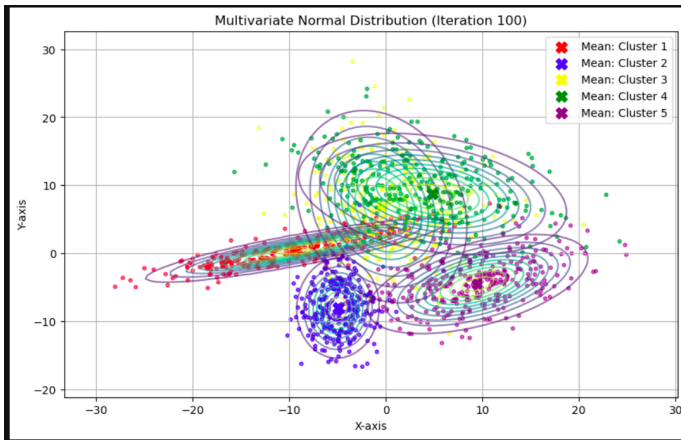


Fig. 5. More challenging data with closer means.

I then tried to challenge the algorithm by providing an incorrect number of clusters. For example, I used data with 5 clusters but instructed the algorithm to cluster it into 4 (Fig. 6) or 6 (Fig. 7) groups.

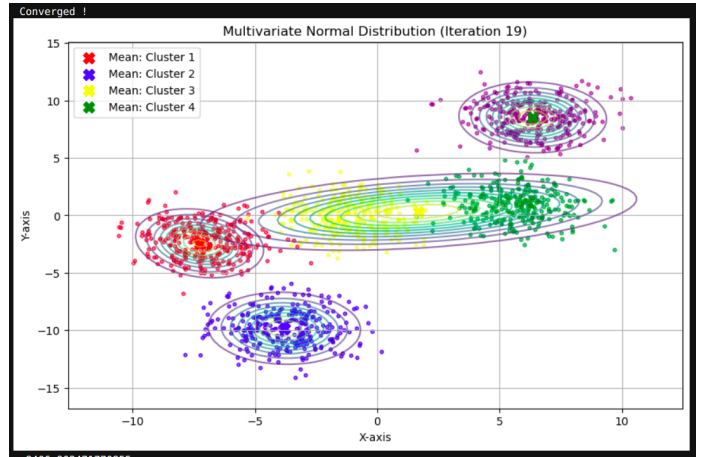


Fig. 6. Algorithm tries to fit the data into 4 clusters.

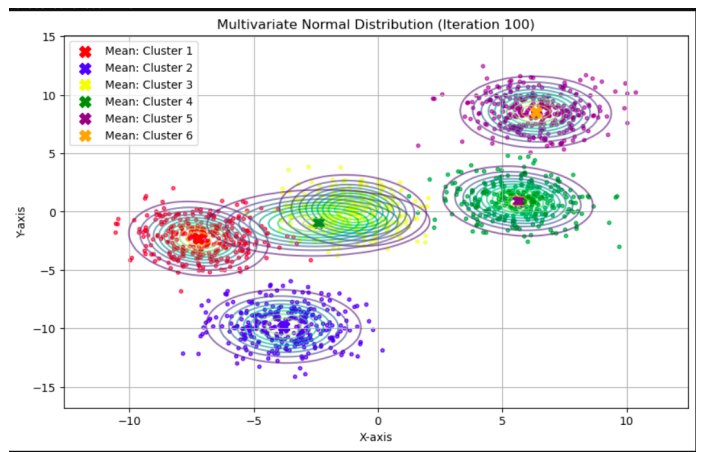


Fig. 7. Algorithm tries to fit the data into 6 clusters.

A. Key Insights

- **Mean proximity:** The closer the means are, the more challenging it is to cluster the data.
- **Covariance similarity:** The closer the covariances are to the identity matrix, the easier it is to cluster the data.
- **Number of clusters:** A higher number of clusters makes it more difficult for the algorithm to cluster effectively.
- **Mismatch between actual and specified number of clusters:** If the algorithm is given an incorrect number of clusters, the clustering quality will decrease.